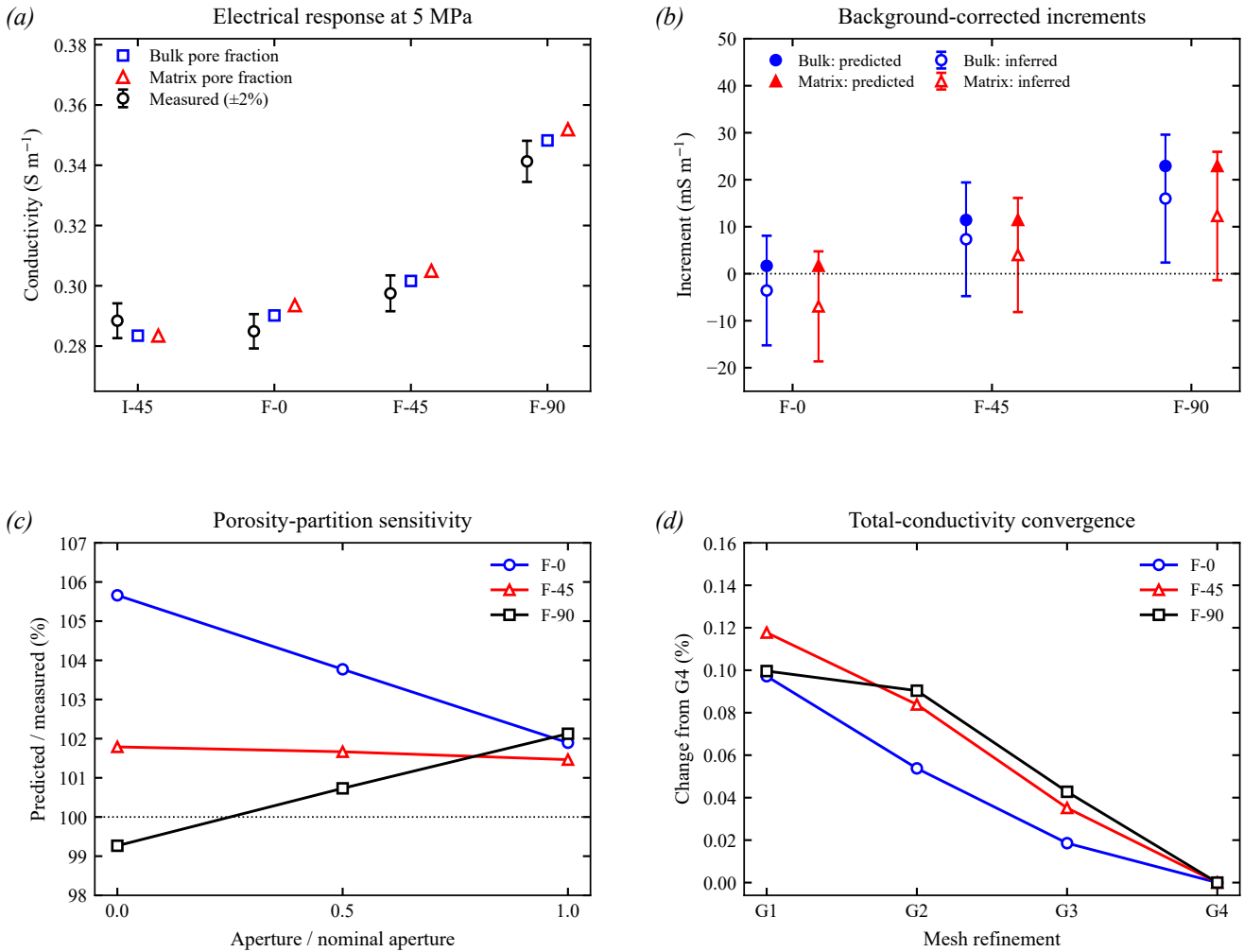

(a)
Electrical response at 5 MPa
Bulk pore fraction
Matrix pore fraction
Measured (±2%)
Conductivity (S m⁻¹)
0.38
0.36
0.34
0.32
0.30
0.28
I-45
F-0
F-45
F-90
(b)
Background-corrected increments
Bulk: predicted
Matrix: predicted
Bulk: inferred
Matrix: inferred
Increment (mS m⁻¹)
50
40
30
20
10
0
−10
−20
F-0
F-45
F-90
(c)
Porosity-partition sensitivity
F-0
F-45
F-90
Predicted / measured (%)
107
106
105
104
103
102
101
100
99
98
0.0
0.5
1.0
Aperture / nominal aperture
(d)
Total-conductivity convergence
F-0
F-45
F-90
Change from G4 (%)
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
G1
G2
G3
G4
Mesh refinement